

Name: Student name

### Quiz 5

Multiple choice questions (1-5): select one answer from the five options.

Fill in the blanks (6,7): write your answer in the space given within the question.

Short answer (8-10): write your answer in the space given below the question.

2 points will be awarded for each correct answer (total 20)

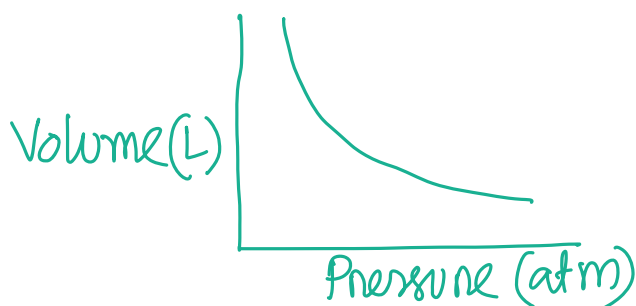
1. A volume of air occupying  $12.0 \text{ dm}^3$  at  $98.9 \text{ kPa}$  is compressed to a pressure of  $119.0 \text{ kPa}$ . The temperature remains constant. What is the new volume?
  - a.  $99.7 \text{ dm}^3$
  - b.  $8.65 \text{ dm}^3$
  - ☒ c.  $9.97 \text{ dm}^3$
  - d.  $10 \text{ dm}^3$
  - e.  $8.89 \text{ dm}^3$
2. If you expect a chemical reaction to produce  $4.38 \text{ dm}^3$  of oxygen,  $\text{O}_2$ , at  $19^\circ\text{C}$  and  $101 \text{ kPa}$ , what will be the volume at  $25^\circ\text{C}$  and  $101 \text{ kPa}$ ?
  - ☒ a.  $4.47 \text{ dm}^3$
  - b.  $5.38 \text{ dm}^3$
  - c.  $4.41 \text{ dm}^3$
  - d.  $3.38 \text{ dm}^3$
  - e.  $44.7 \text{ dm}^3$
3. Which statement is wrong?
  - ☒ a. In Boyle's law, only temperature needs to be constant
  - b. Both pressure and amount of gas are constant in Charles's law
  - c. At a fixed temperature for a given amount of gas, the product of pressure and volume will be constant
  - d. At STP, molar gas volume is found to be  $22.4 \text{ dm}^3$
  - e. From ideal gas law,  $P/n = (R/V) \cdot T$
4. How many grams of oxygen,  $\text{O}_2$ , are there in a  $50.0 \text{ L}$  gas cylinder at  $21^\circ\text{C}$  when the oxygen pressure  $15.7 \text{ atm}$ ?
  - ☒ a.  $1.04 \times 10^3 \text{ g O}_2$
  - b.  $1.78 \times 10^3 \text{ g O}_2$
  - c.  $3.24 \times 10^3 \text{ g O}_2$
  - d.  $10.4 \times 10^4 \text{ g O}_2$
  - e.  $1.39 \times 10^2 \text{ g O}_2$
5. A sample of a gaseous substance at  $25^\circ\text{C}$  and  $0.862 \text{ atm}$  has a density of  $2.26 \text{ g/L}$ . What is the molecular mass of the substance?
  - ☒ a.  $154 \text{ g/mol}$
  - b.  $77 \text{ g/mol}$
  - c.  $130 \text{ g/mol}$
  - d.  $127 \text{ g/mol}$
  - e.  $127.8 \text{ g/mol}$

6. Gaseous diffusion is the process whereby a gas spreads out through another gas to occupy the space uniformly.
7. Graham's law of effusion states the relation between the effusion of a gas to its molecular mass.

8. Write the van der Waals equation for the pressure correction of an ideal gas.

$$P(\text{actual}) = P(\text{ideal}) - \frac{n^2 a}{V^2}$$

9. Draw the plot of volume vs. pressure for a sample gas at a temperature?



10. What is partial pressure?

The pressure exerted by a particular gas in a mixture.